Experiment No. 1

Aim: To study different types of wireless Technology

Wireless Technology

1. Wi-Fi (Wireless Fidelity)

- **Range of Communication**: Typically up to 100 meters (indoors). The range can extend with the use of Wi-Fi extenders or routers supporting higher frequencies like 5 GHz.
- **Cost**: Moderate. Wi-Fi routers and devices are widely available at various price points.
- **Propagation Delay**: Moderate. Wi-Fi operates on the 2.4 GHz and 5 GHz bands, which are relatively less prone to significant delays compared to lower-frequency systems like LoRa.
- **Power Consumption**: Moderate to high. Devices like smartphones, laptops, and routers consume a considerable amount of power.
- **Throughput**: High. Wi-Fi can support speeds ranging from 72 Mbps (Wi-Fi 4) up to 9.6 Gbps (Wi-Fi 6).
- **Fanout**: High. Wi-Fi can handle numerous devices simultaneously (hundreds to thousands in a managed environment like a Wi-Fi 6 network).
- **Applications**: Home/office networking, public Wi-Fi, streaming, smart home devices.

2. Bluetooth

- **Range of Communication**: Short-range, typically up to 100 meters (for Bluetooth 5.x), but often much shorter (10 meters) in practical use.
- **Cost**: Low. Bluetooth chips are inexpensive, and devices supporting Bluetooth are common.
- **Propagation Delay**: Low. Bluetooth has low latency, ideal for applications like audio streaming.
- **Power Consumption**: Low. Bluetooth Low Energy (BLE) is designed to use minimal power, making it suitable for IoT devices.
- **Throughput**: Low. Bluetooth throughput typically ranges from 1 Mbps to 2 Mbps (BLE can go higher, but still not comparable to Wi-Fi).
- **Fanout**: Moderate. Bluetooth can connect several devices in a network, but it’s not ideal for large-scale communication compared to Wi-Fi or Zigbee.
- **Applications**: Wireless audio (headphones, speakers), health devices (fitness trackers), IoT devices, peripherals (keyboards, mice).

3. Zigbee

- **Range of Communication**: Short to medium range (10-100 meters), depending on the environment.
- **Cost**: Low. Zigbee modules are inexpensive and widely used in IoT devices.
- **Propagation Delay**: Low to moderate. Zigbee is designed for low-latency, low-power applications.
- **Power Consumption**: Very low. Zigbee is optimized for low power usage, often running on small batteries for extended periods (years in some cases).
- **Throughput**: Low. Zigbee offers a maximum throughput of around 250 kbps.
- **Fanout**: High. Zigbee supports mesh networking, which allows many devices to communicate with one another, enabling large networks in IoT scenarios.
- **Applications**: Home automation, industrial IoT, smart energy, healthcare, building management.

4. LoRa (Long Range)

- **Range of Communication**: Very long-range, up to 15-20 km in rural areas and several kilometers in urban environments.
- **Cost**: Low. LoRa devices are relatively inexpensive and used in applications like smart agriculture and IoT.
- **Propagation Delay**: High. Due to the low-frequency signals used (typically around 868 MHz and 915 MHz), LoRa can experience higher propagation delay.
- **Power Consumption**: Very low. LoRa devices are designed for low power consumption, which allows them to run for years on a single battery.
- **Throughput**: Low. LoRa supports data rates from 0.3 kbps to 27 kbps, depending on the settings.
- **Fanout**: Moderate. LoRa supports a limited number of devices per gateway in a wide area network setup, but it’s not as scalable as cellular networks.
- **Applications**: Smart cities, agriculture, environmental monitoring, asset tracking, long-range IoT networks.

5. 5G (Fifth Generation)

- **Range of Communication**: Moderate. The range varies greatly depending on the frequency used, from several kilometers in low-band (Sub-6 GHz) to a few hundred meters in high-band (mmWave).
- **Cost**: High. Infrastructure costs for 5G are significant, and devices supporting 5G are more expensive than those for 4G or Wi-Fi.
- **Propagation Delay**: Very low. 5G promises ultra-low latency with a typical delay of around 1 ms.
- **Power Consumption**: Moderate to high. 5G devices require more power than 4G or Wi-Fi but are optimized for high-performance, low-latency connections.
- **Throughput**: Very high. 5G can theoretically reach up to 20 Gbps, with real-world speeds typically ranging from 1 to 10 Gbps.
- **Fanout**: Very high. 5G is designed to support massive numbers of connected devices (including IoT) and large-scale fanout.

- **Applications**: Mobile broadband, IoT, autonomous vehicles, smart manufacturing, VR/AR, telemedicine.

6. NFC (Near Field Communication)

- **Range of Communication**: Very short (typically less than 10 cm).
- **Cost**: Very low. NFC chips are extremely inexpensive, and they are often integrated into smartphones, smart cards, and payment systems.
- **Propagation Delay**: Very low. NFC has near-instantaneous communication with minimal delay.
- **Power Consumption**: Low. NFC typically uses passive power from the reader device, so the powered device is usually the one consuming energy.
- **Throughput**: Very low. NFC supports data transfer rates up to 424 kbps.
- **Fanout**: Low. NFC is designed for point-to-point communication, typically between two devices only.
- **Applications**: Mobile payments, access control, ticketing, authentication, data sharing.

7. Infrared (IR)

- **Range of Communication**: Short range, typically up to 10 meters.
- **Cost**: Low. IR transmitters and receivers are very inexpensive.
- **Propagation Delay**: Low. IR signals travel at the speed of light with negligible delay.
- **Power Consumption**: Low. IR devices consume very little power, especially in consumer electronics like remote controls.
- **Throughput**: Low. IR supports a data rate up to 4 Mbps, but this is typically lower than other wireless technologies.
- **Fanout**: Low. IR is mostly used for one-to-one communication, such as in remote controls.
- **Applications**: Remote controls, wireless data transfer, motion detection, medical thermometers, robotics.

8. Satellite Communication

- **Range of Communication**: Very long range. Satellite communication can cover the entire Earth, making it ideal for global communication.
- **Cost**: High. Setting up and maintaining satellite infrastructure is expensive, and devices can also be costly.
- **Propagation Delay**: High. The long distance between Earth and satellites results in noticeable latency, typically 500-600 ms for geostationary satellites.
- **Power Consumption**: High. Satellite communication requires significant power for both the satellite and the ground stations.
- **Throughput**: Moderate to high. Throughput can vary depending on the satellite system; typical commercial satellite services provide speeds from a few Mbps to several hundred Mbps.

- **Fanout**: High. Satellites can handle large-scale communication for a vast number of devices globally.
- **Applications**: Global communication, broadcasting, weather forecasting, GPS, search and rescue.

9. Cellular Communication (4G/5G/Other Generations)

- **Range of Communication**: Moderate. Cellular networks can cover vast areas, typically from several kilometers in rural areas to a few hundred meters in urban areas, depending on the generation (4G, 5G).
- **Cost**: High. Infrastructure costs are substantial, and mobile data plans can add to the cost for end users.
- **Propagation Delay**: Moderate to low. 4G has latencies of 30-50 ms, while 5G can lower this to sub-10 ms in ideal conditions.
- **Power Consumption**: Moderate to high. Cellular devices like smartphones consume a decent amount of power, especially on 4G/5G.
- **Throughput**: High. 4G typically offers speeds up to 100 Mbps, while 5G can offer speeds from 1 to 20 Gbps, depending on network conditions.
- **Fanout**: Very high. Cellular networks are built for large-scale deployment, serving millions of users with high fanout.
- **Applications**: Mobile phones, mobile broadband, smart cities, connected vehicles, industrial automation, telemedicine.

Technology	Range	Cost	Propagation Delay	Power Consumption	Throughput	Fanout
Wi-Fi	100 meters	Moderate	Moderate	Moderate to high	High (up to 9.6 Gbps)	High
Bluetooth	100 meters	Low	Low	Low	Low (up to 2 Mbps)	Moderate
Zigbee	10-100 meters	Low	Low to moderate	Very low	Low (250 kbps)	High
LoRa	15-20 km	Low	High	Very low	Low (up to 27 kbps)	Moderate
5G	Several km	High	Very low	Moderate to high	Very high (up to 20 Gbps)	Very High
NFC	<10 cm	Very low	Very low	Low	Very low (424 kbps)	Low

Infrared	10 meters	Low	Low	Low	Low (up to 4 Mbps)	Low
Satellite	Global coverage	High	High	High	Moderate to high	High
Cellular (4G/5G)	Several km	High	Moderate to low	Moderate to high	High (4G: 100 Mbps, 5G: up to 20 Gbps)	Very High